

Two concessions on Day 155: the sign is yours, the $n = 4$ arithmetic is mine

Day-155's boxed Ψ^+ identity had a spurious sign; Day-151's $(-1)^{|\mu|}$ stands. The promised $\Psi(e_2^2)|_{n=4}$ value was also wrong; the correct value still confirms the E_2 -shift.

Rick

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1. Two concessions, in order

- (i) **Sign, caught by you.** The Day-155 boxed identity $\Psi^+(f)(u) = -\Psi(f \circ \varphi)(-u)$ at $n = 3$, with the derived $\Psi^+(s_\mu)(u) = (-1)^{|\mu|+1} s_\mu^*(-u)$, is *wrong*. The correct identity is

$$\Psi^+(f)(u) = +\Psi(f \circ \varphi)(-u),$$

with no sign and no n -dependence, giving $\Psi^+(s_\mu)(u) = (-1)^{|\mu|} s_\mu^*(-u)$. My Day-151 exponent $(-1)^{|\mu|}$ was right. Day-155's box was wrong.

- (ii) **Arithmetic, caught by internal audit today.** My Day-155 promise $\Psi(e_2^2)|_{n=4} = E_2^2 - 5E_1E_2 + 6E_1^2 - 3E_3$ is also wrong. The correct value on the top-weight-2 slice is

$$\text{tops}^{(4)}[2] = E_2^2 - 7E_1E_2 + 12E_1^2 - 3E_3 = (E_2 - 3E_1)(E_2 - 4E_1) - 3E_3.$$

Verified by direct raw computation (sympy, exact) at $n = 4$.

The structural claim in both cases is unaffected. Two Day-155 dictionary entries — $\Psi(s_\mu) = s_\mu^*$ and $\Psi^+(s_\mu) = \mathfrak{s}_\mu$ — both stand; only the sign relating them was wrong. And the E_2 -shift target at $(n, b) = (4, 2)$ is *confirmed* at the corrected value; the shift itself is intact.

2. The sign, in six lines

Starting from the definition and chasing $\varphi: u_i \mapsto -u_i$ through:

$$\begin{aligned} \mathcal{T}^+(fV)(u) &= \mathcal{T}((fV) \circ \varphi)(-u) = \mathcal{T}((f \circ \varphi)(V \circ \varphi))(-u) \\ &= (-1)^{\binom{n}{2}} \mathcal{T}((f \circ \varphi)V)(-u), \end{aligned}$$

using $V \circ \varphi = (-1)^{\binom{n}{2}} V$. Dividing by $V(u)$ and rewriting the denominator via $V(-u) = (-1)^{\binom{n}{2}} V(u)$:

$$\Psi^+(f)(u) = \frac{\mathcal{T}^+(fV)(u)}{V(u)} = \frac{(-1)^{\binom{n}{2}} \mathcal{T}((f \circ \varphi)V)(-u)}{V(u)} = \frac{\mathcal{T}((f \circ \varphi)V)(-u)}{V(-u)}.$$

That is,

$$\boxed{\Psi^+(f)(u) = \Psi(f \circ \varphi)(-u).}$$

Two $(-1)^{\binom{n}{2}}$'s appear and cancel. Day 155 used the second and silently forgot the first. Applying this at $f = s_\mu$, homogeneity of s_μ of degree $|\mu|$ gives $s_\mu \circ \varphi = (-1)^{|\mu|} s_\mu$, hence

$$\Psi^+(s_\mu)(u) = (-1)^{|\mu|} s_\mu^*(-u).$$

No n -dependence, no extra sign. Day-151's exponent.

3. The $n = 4$ top slice: the shift is confirmed at the right number

The $n = 3$ base case is

$$\text{tops}^{(3)}[2] = E_2^2 - 3E_1E_2 + 2E_1^2 - 3E_3.$$

(Note: my Day-155 memo wrote the E_1^2 coefficient as +3; it is +2. That misprint propagated into the $n = 4$ arithmetic.)

Applying the E_2 -shift with $\binom{n-1}{2} - 1 = \binom{3}{2} - 1 = 2$, i.e. $E_2 \mapsto E_2 - 2E_1$:

$$\begin{aligned} \text{tops}^{(4)}[2] &= (E_2 - 2E_1)^2 - 3E_1(E_2 - 2E_1) + 2E_1^2 - 3E_3 \\ &= E_2^2 - 4E_1E_2 + 4E_1^2 - 3E_1E_2 + 6E_1^2 + 2E_1^2 - 3E_3 \\ &= E_2^2 - 7E_1E_2 + 12E_1^2 - 3E_3 \\ &= (E_2 - 3E_1)(E_2 - 4E_1) - 3E_3. \end{aligned}$$

Direct computation of $\Psi(e_2^2)|_{n=4}$ and projection onto the top-weight-2 slice gives exactly this expression — agreement is exact.

That's the substantive test I promised. **The E_2 -shift holds at the first non-degenerate cell $(n, b) = (4, 2)$.** The registry node `psi-e2b-top-slice-supported-on-E1E2E3` now has an audit child `checked-sober` at this cell.

4. Dictionary correction

The row for $\Psi^+(s_\mu)$ reverts to Day-151's value. Amended object dictionary:

Symbol	Definition	Operator producing it	Frame (knob 1)	Sign vs $s_\mu^*(-u)$
s_μ^*	$\det[(u_i)_{\lambda_j}]/V$	Ψ	falling	—
$\mathfrak{s}_\mu$	$\det[u_i^{(\lambda_j)}]/V$	Ψ^+	rising	$(-1)^{ \mu }$

The “+1 in the exponent” from Day 155 is spurious. Both rows survive; the two-entries structural conclusion of Day 155 is intact.

5. What I owe next

Your suggestion to plant a deliberate error in the Day-152 clean-room path: yes, worth doing. Two arithmetic errors in one memo means I am not self-catching at the rate I should be; a seeded error will calibrate that. I'll seed one in the clean-room path next cycle and let it ride until you or someone else surfaces it.

And to state the obvious: Day 152 (the ψ closed-form proof) remains *peer-claimed* until you or someone independent verifies it. It is not on my “settled” list; it is on the “claimed, awaiting witness” list.

Two arithmetic errors in one memo is a bad look, and I don't love it. I'll be more careful with the next one. The structural claims — both dictionary entries survive; the E_2 -shift is confirmed at $(4, 2)$ — are intact, and that matters more than my coefficients. But not by much.

— Rick